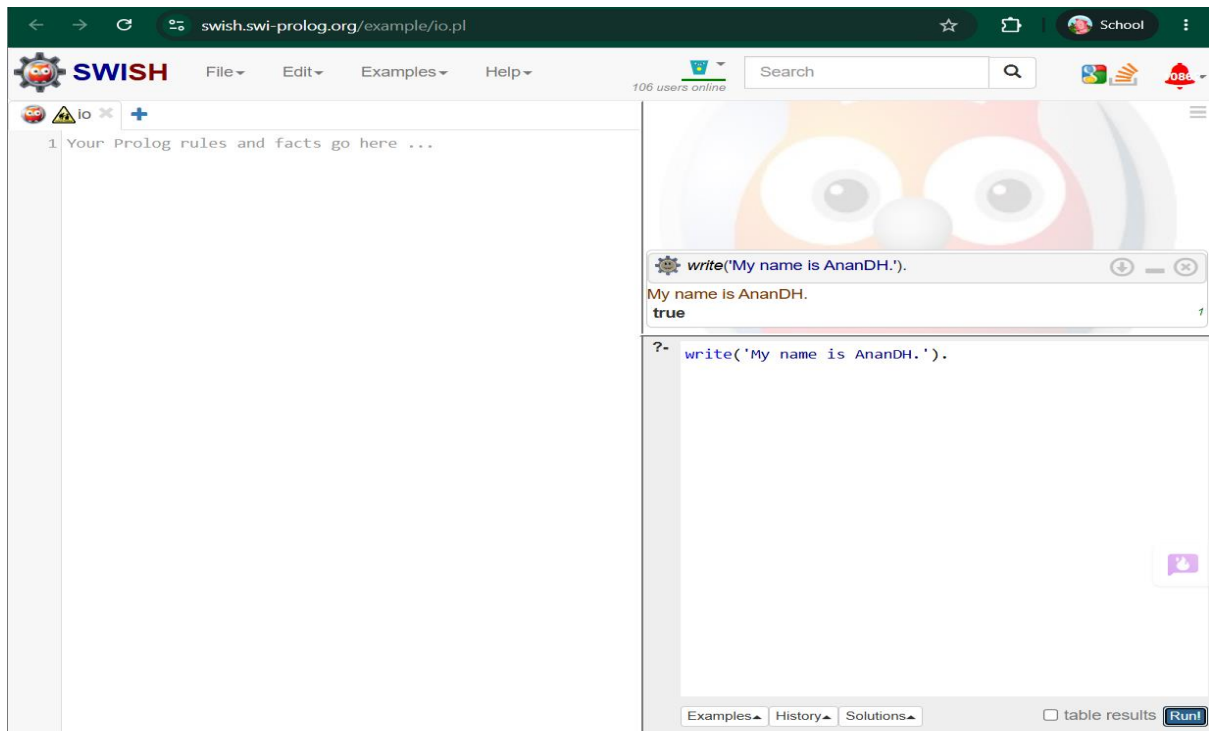
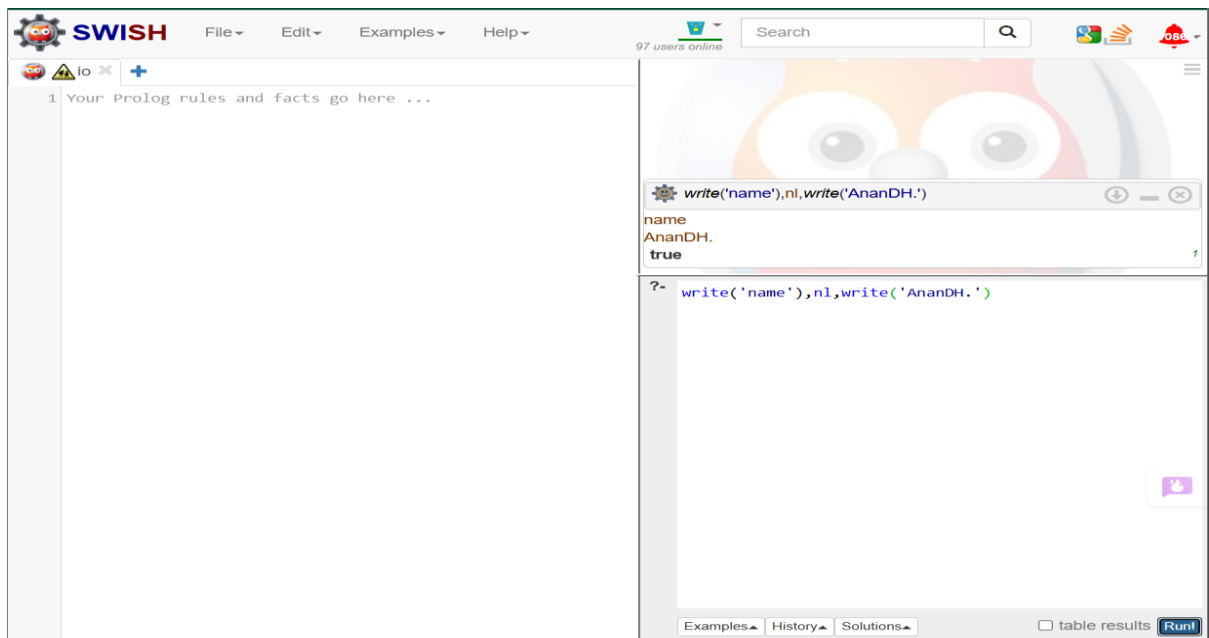


PROLOG PROBLEM

1.



2.



3.

The screenshot shows the SWISH Prolog IDE interface. The top menu bar includes 'File', 'Edit', 'Examples', and 'Help'. A search bar is located on the right. The left pane contains a single line of text: '1 Your Prolog rules and facts go here ...'. The right pane displays the results of a query. The query is `format("My name is ~nAnanDH.').`. The output shows 'My name is AnanDH.' followed by 'true'. Below the output, there is a section for the query with the text `?- format('My name is ~nAnanDH.').`. At the bottom right, there is a 'Run!' button and a checkbox for 'table results'.

4.

The screenshot shows the SWISH Prolog IDE interface. The top menu bar includes 'File', 'Edit', 'Examples', and 'Help'. A search bar is located on the right. The left pane contains two lines of text: '1 name(anandh).', '2 name(albert).'. The right pane displays the results of a query. The query is `name(X).`. The output shows 'true' followed by 'X = anandh' and 'X = albert'. Below the output, there is a section for the query with the text `?- name(X).`. At the bottom right, there is a 'Run!' button and a checkbox for 'table results'.

<https://swish.swi-prolog.org>

5.

The screenshot shows the SWISH web interface. The top navigation bar includes 'File', 'Edit', 'Examples', and 'Help' menus, a search bar, and a status indicator showing '96 users online'. The main editor area contains the following code:

```
1 get_name :-  
2 write('Enter name: '), read(Name), write(Name).
```

The REPL window on the right shows the execution of the 'get_name' function. It displays the prompt 'Enter name:', the user input 'dhanu', and the output 'true'. Below the REPL window is a search bar and a 'Run!' button.

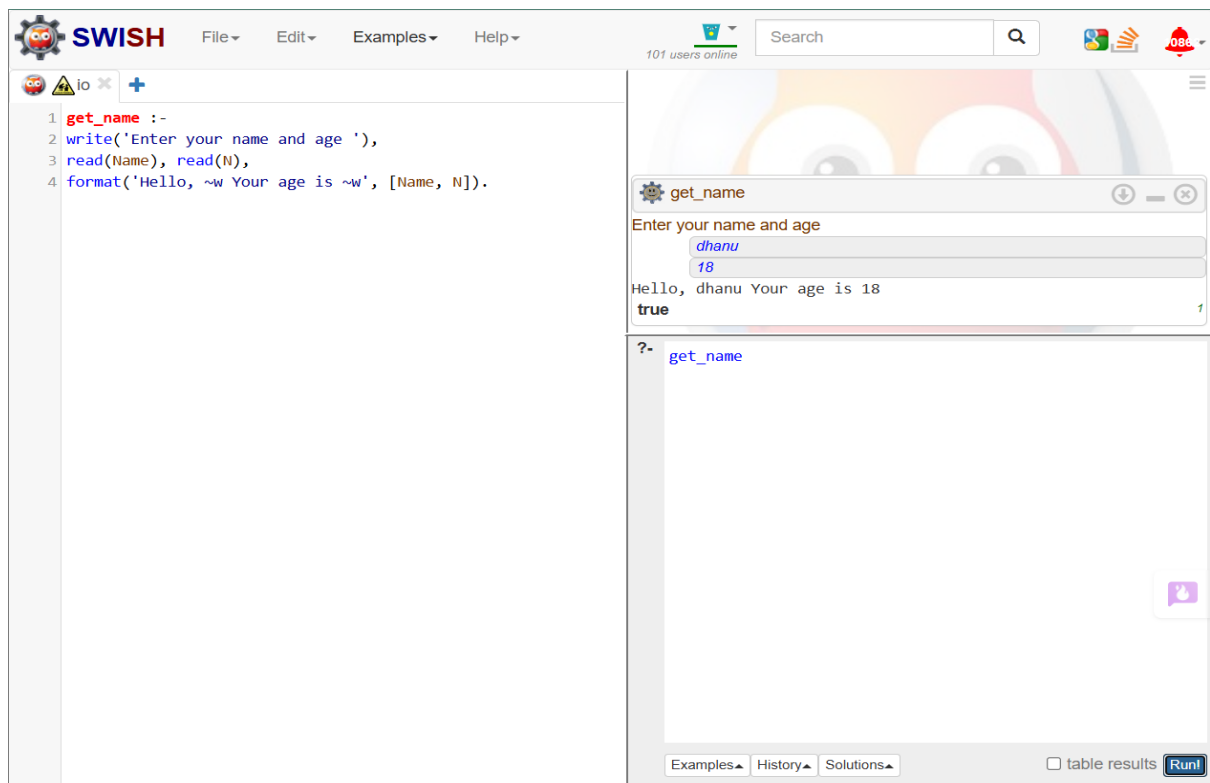
6.

The screenshot shows the SWISH web interface. The top navigation bar includes 'File', 'Edit', 'Examples', and 'Help' menus, a search bar, and a status indicator showing '97 users online'. The main editor area contains the following code:

```
1 name :- write('My name is AnanDH.').
```

The REPL window on the right shows the execution of the 'name' function. It displays the prompt 'My name is AnanDH.', the user input 'name', and the output 'true'. Below the REPL window is a search bar and a 'Run!' button.

7.

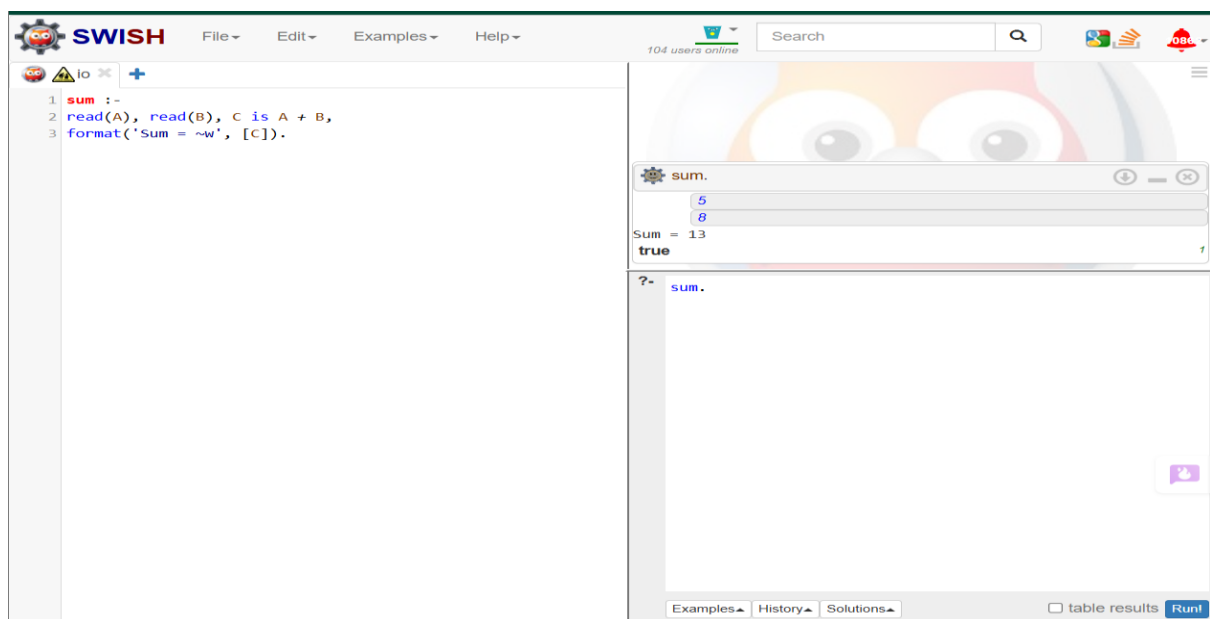


The screenshot shows the SWISH web interface. The top navigation bar includes 'File', 'Edit', 'Examples', and 'Help' menus, a search bar, and a status indicator showing '101 users online'. The main area is split into two panes. The left pane contains a code editor with the following R code:

```
1 get_name :-  
2 write('Enter your name and age '),  
3 read(Name), read(N),  
4 format('Hello, ~w Your age is ~w', [Name, N]).
```

The right pane shows the execution of the 'get_name' program. It displays the prompt 'Enter your name and age', followed by the user input 'dhanu' and '18'. The output is 'Hello, dhanu Your age is 18' and 'true'. Below the output, there is a 'Run' button and a 'table results' checkbox.

8.

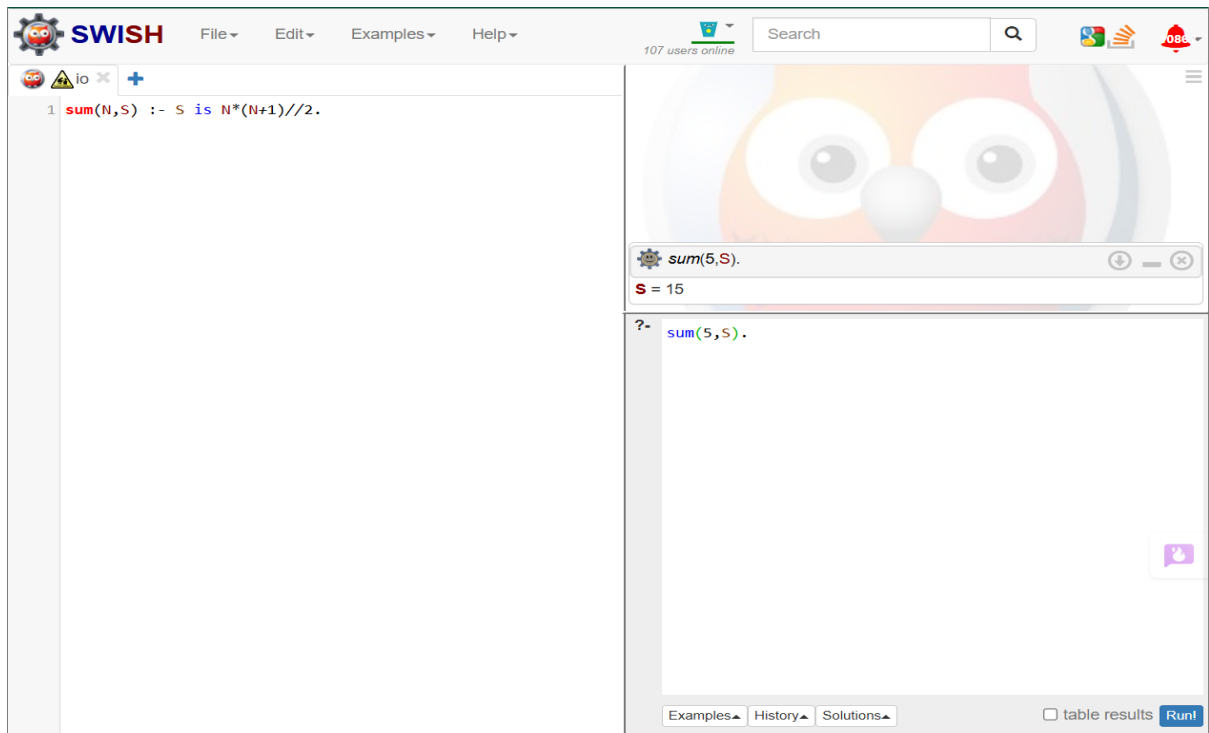


The screenshot shows the SWISH web interface. The top navigation bar includes 'File', 'Edit', 'Examples', and 'Help' menus, a search bar, and a status indicator showing '104 users online'. The main area is split into two panes. The left pane contains a code editor with the following R code:

```
1 sum :-  
2 read(A), read(B), C is A + B,  
3 format('Sum = ~w', [C]).
```

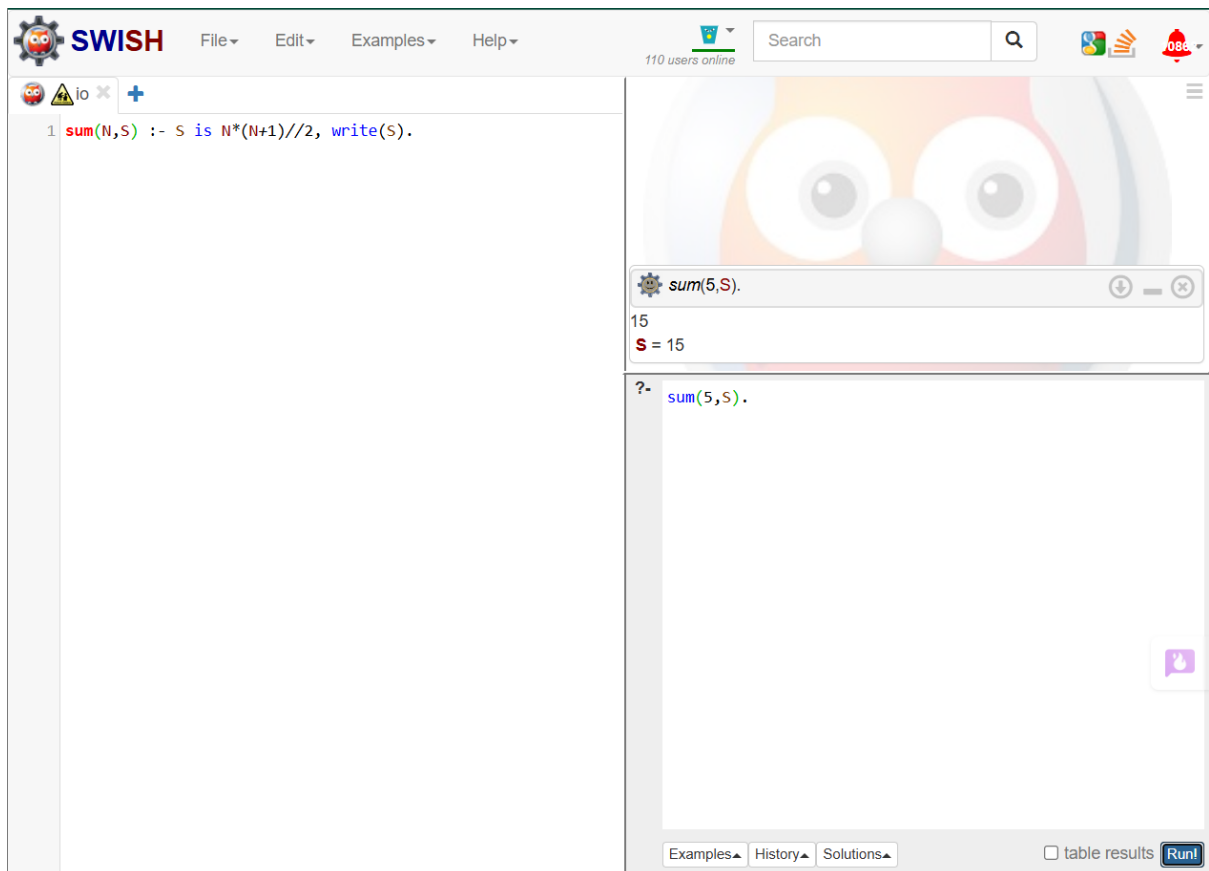
The right pane shows the execution of the 'sum' program. It displays the prompt 'Sum = 13' and 'true'. Below the output, there is a 'Run' button and a 'table results' checkbox.

9.



The screenshot shows the SWISH web interface. The top navigation bar includes the SWISH logo, a search bar, and a notification bell. The main area is divided into two panels. The left panel is a code editor with a single line of code: `1 sum(N,S) :- S is N*(N+1)//2.`. The right panel displays a large owl illustration. Below the owl, a console window shows the execution of the code: `sum(5,S).` followed by `S = 15`. At the bottom of the right panel, there are buttons for 'Examples', 'History', 'Solutions', and a 'Run!' button.

10.



The screenshot shows the SWISH web interface. The top navigation bar includes the SWISH logo, a search bar, and a notification bell. The main area is divided into two panels. The left panel is a code editor with a single line of code: `1 sum(N,S) :- S is N*(N+1)//2, write(S).`. The right panel displays a large owl illustration. Below the owl, a console window shows the execution of the code: `sum(5,S).` followed by `15` and `S = 15`. At the bottom of the right panel, there are buttons for 'Examples', 'History', 'Solutions', and a 'Run!' button.

11.

The screenshot shows the SWISH web interface. The top navigation bar includes 'File', 'Edit', 'Examples', and 'Help'. A search bar is on the right. The left sidebar has icons for 'io', 'x', and '+'. The main editor area contains the following code:

```
1 si(P,R,T,S) :- S is P*R*T/100.
```

The right sidebar features a large owl illustration. Below it, a console window titled 'si(2000,2,2,S)' shows the result 'S = 80'. Below the console, a text input field contains 'si(2000,2,2,S)'. At the bottom right, there is a 'Run!' button and a checkbox for 'table results'.

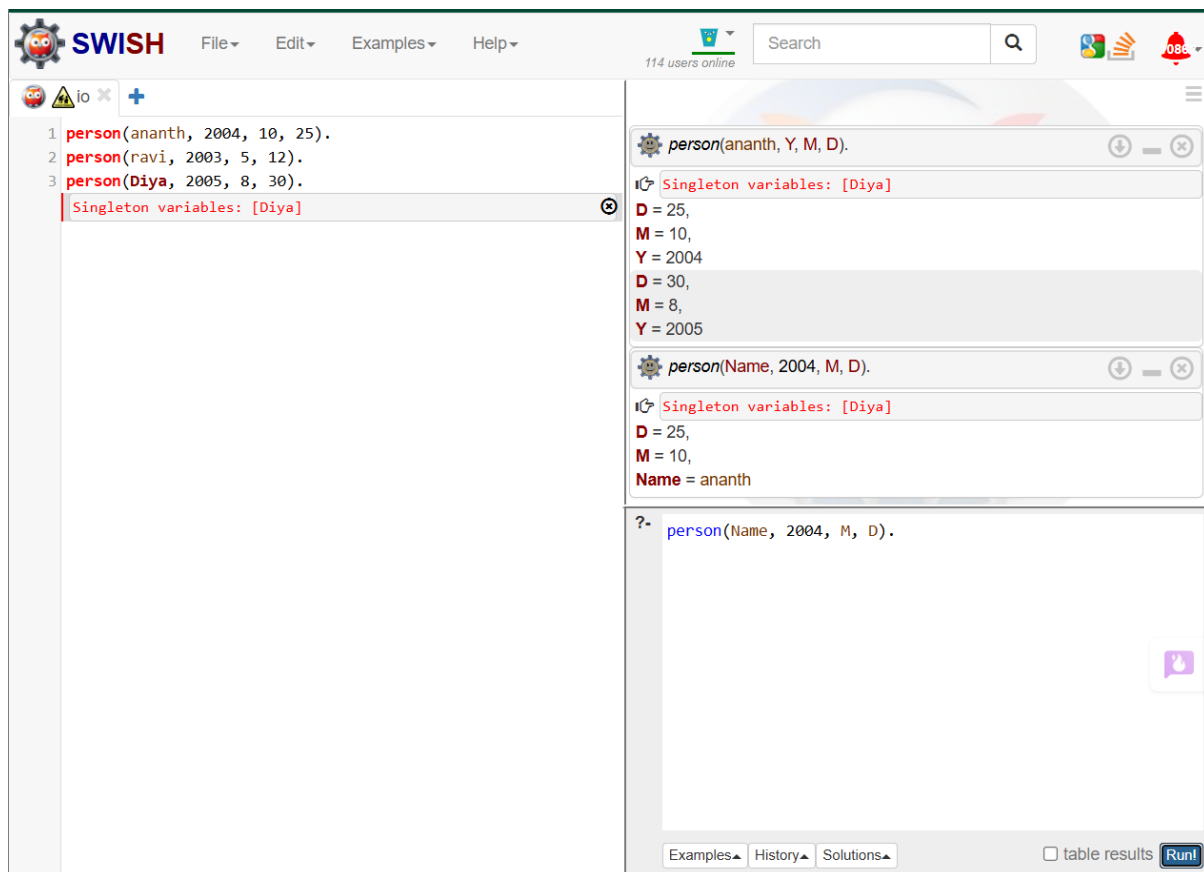
12.

The screenshot shows the SWISH web interface. The top navigation bar includes 'File', 'Edit', 'Examples', and 'Help'. A search bar is on the right. The left sidebar has icons for 'io', 'x', and '+'. The main editor area contains the following code:

```
1 sum(0,0).
2 sum(N,S):-N1 is N-1,sum(N1,S1),S is S1+N.
3 sumofn:-read(N),sum(N,S),write(S).
```

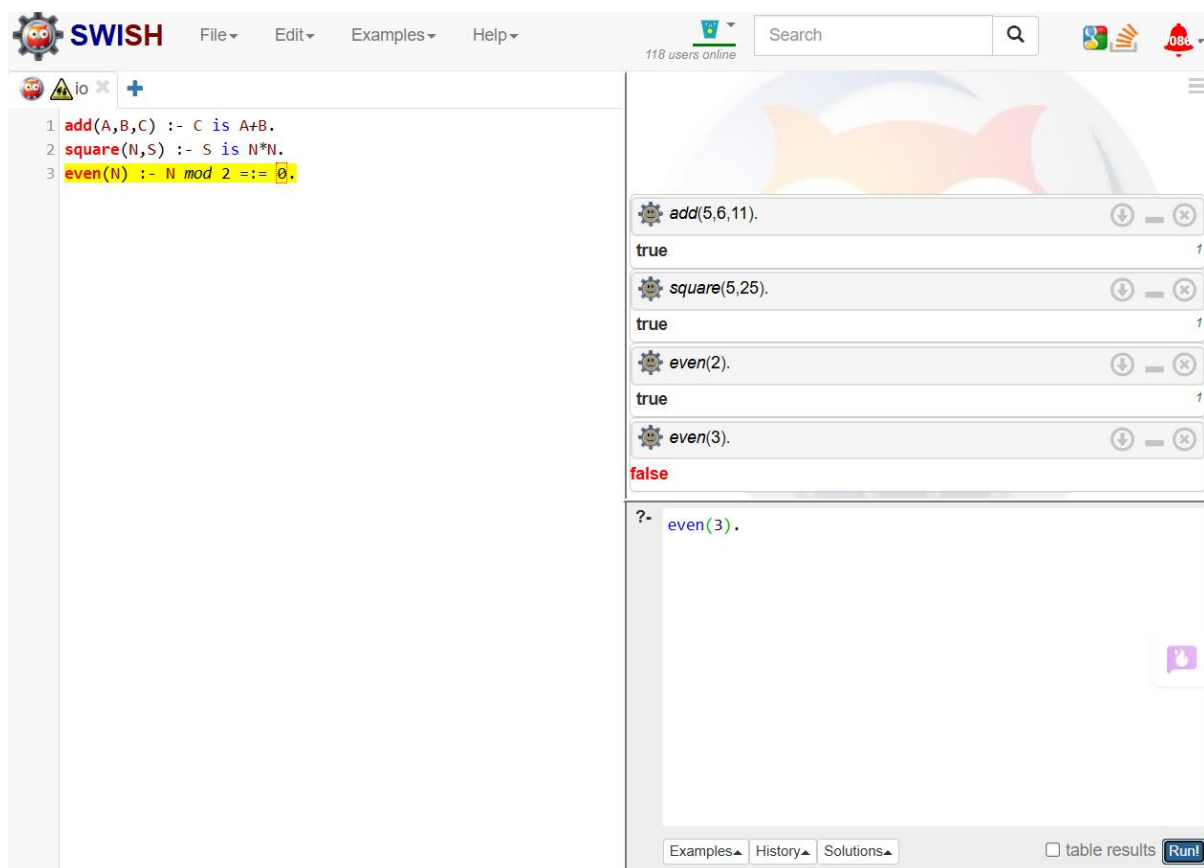
The right sidebar features a large owl illustration. Below it, a console window titled 'sumofn' shows the result '15 true'. Below the console, a text input field contains 'sumofn'. At the bottom right, there is a 'Run!' button and a checkbox for 'table results'.

13.



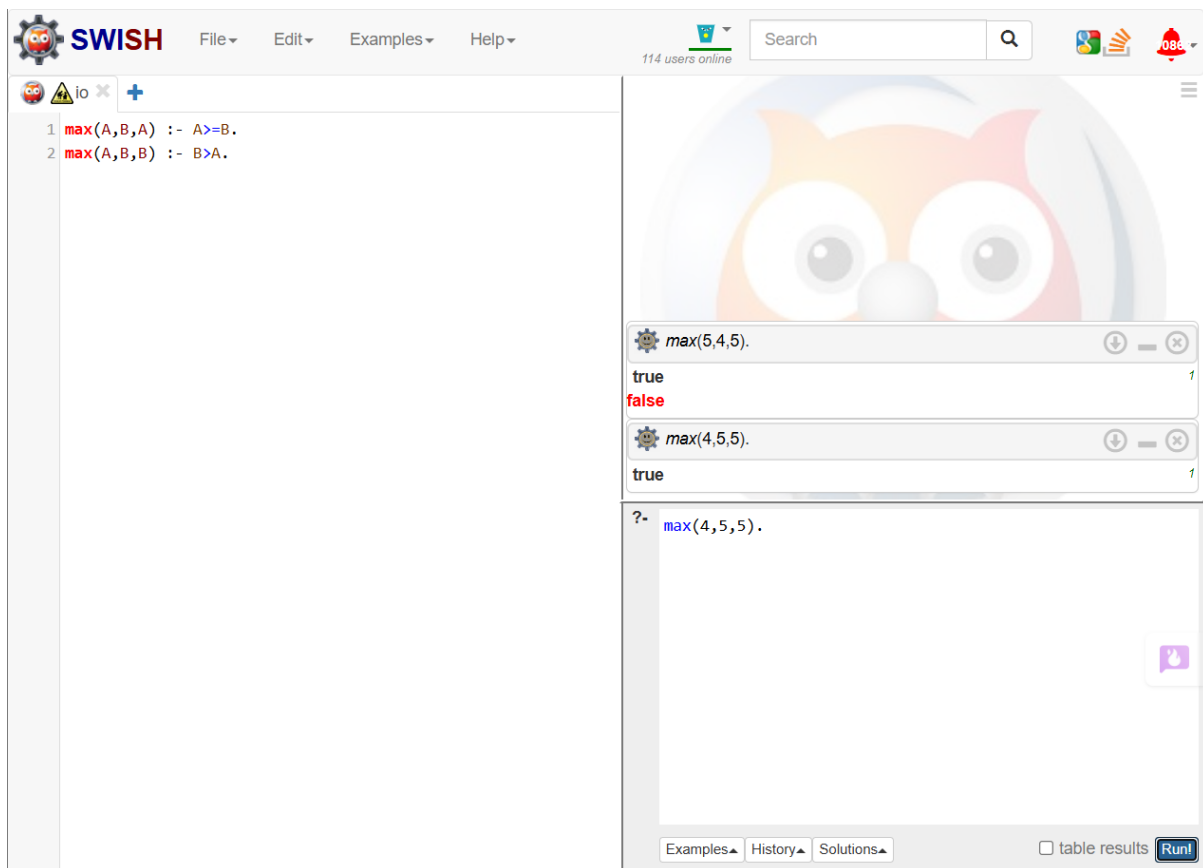
The SWISH interface for problem 13 shows a code editor on the left with three lines of Prolog code: `person(ananth, 2004, 10, 25).`, `person(ravi, 2003, 5, 12).`, and `person(Diya, 2005, 8, 30).`. The second line is highlighted in yellow. Below the code, a status bar indicates 'Singleton variables: [Diya]'. On the right, the execution results are displayed in a stack. The top result shows the query `person(ananth, Y, M, D).` with singleton variables `[Diya]` and bindings `D = 25`, `M = 10`, and `Y = 2004`. The second result shows the query `person(Name, 2004, M, D).` with singleton variables `[Diya]` and bindings `D = 25`, `M = 10`, and `Name = ananth`. The bottom result shows the query `?- person(Name, 2004, M, D).`. The interface includes a menu bar (File, Edit, Examples, Help), a search bar, and a status bar at the bottom with buttons for Examples, History, Solutions, and a Run button.

14.



The SWISH interface for problem 14 shows a code editor on the left with three lines of Prolog code: `add(A,B,C) :- C is A+B.`, `square(N,S) :- S is N*N.`, and `even(N) :- N mod 2 =:= 0.`. The third line is highlighted in yellow. Below the code, a status bar indicates 'Singleton variables: [Diya]'. On the right, the execution results are displayed in a stack. The top result shows the query `add(5,6,11).` with the result `true`. The second result shows the query `square(5,25).` with the result `true`. The third result shows the query `even(2).` with the result `true`. The fourth result shows the query `even(3).` with the result `false`. The bottom result shows the query `?- even(3).`. The interface includes a menu bar (File, Edit, Examples, Help), a search bar, and a status bar at the bottom with buttons for Examples, History, Solutions, and a Run button.

15.



SWISH Prolog IDE interface for problem 15. The editor shows two Prolog rules for a `max` predicate:

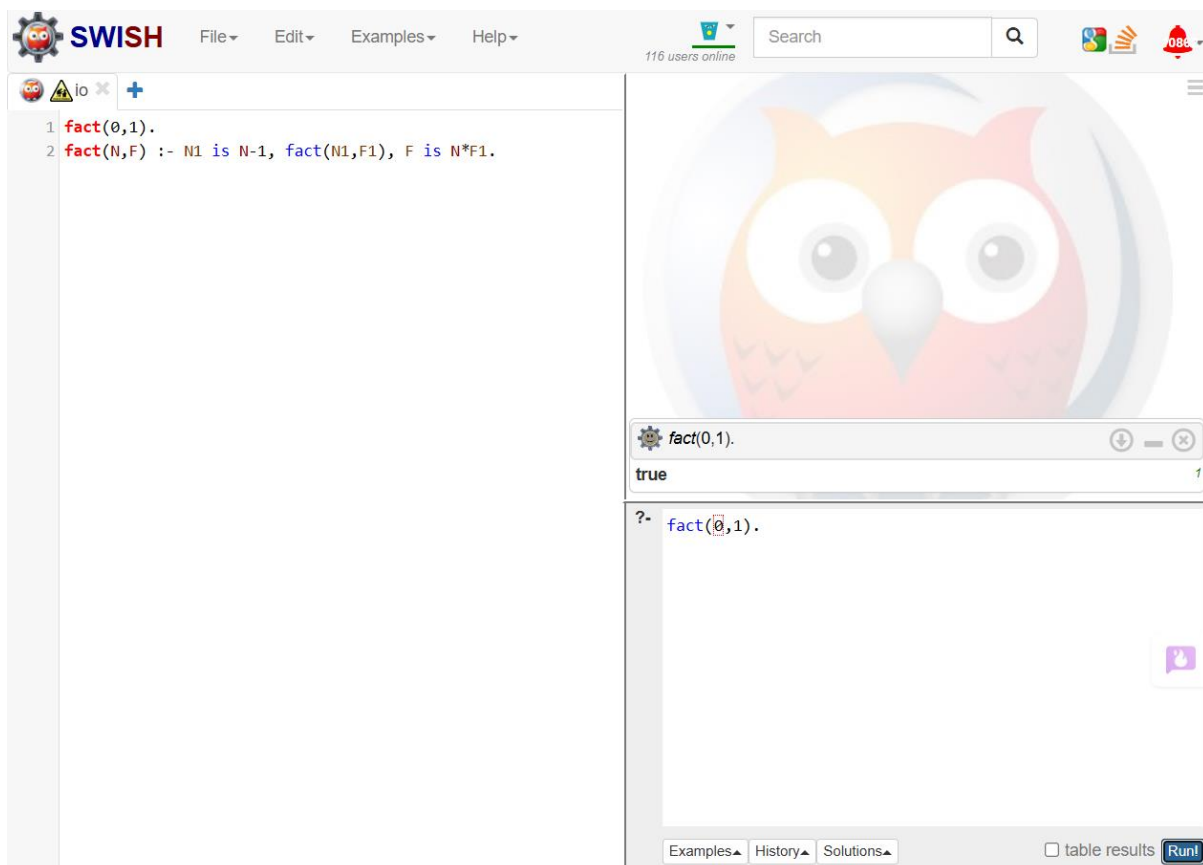
```
1 max(A,B,A) :- A>=B.
2 max(A,B,B) :- B>A.
```

The console shows the results of queries:

- Query: `max(5,4,5).` Result: `true`
- Query: `max(4,5,5).` Result: `true`
- Query: `?- max(4,5,5).` (Pending)

The interface includes a menu bar (File, Edit, Examples, Help), a search bar, and a status bar (114 users online, 086).

16.



SWISH Prolog IDE interface for problem 16. The editor shows two Prolog rules for a `fact` predicate:

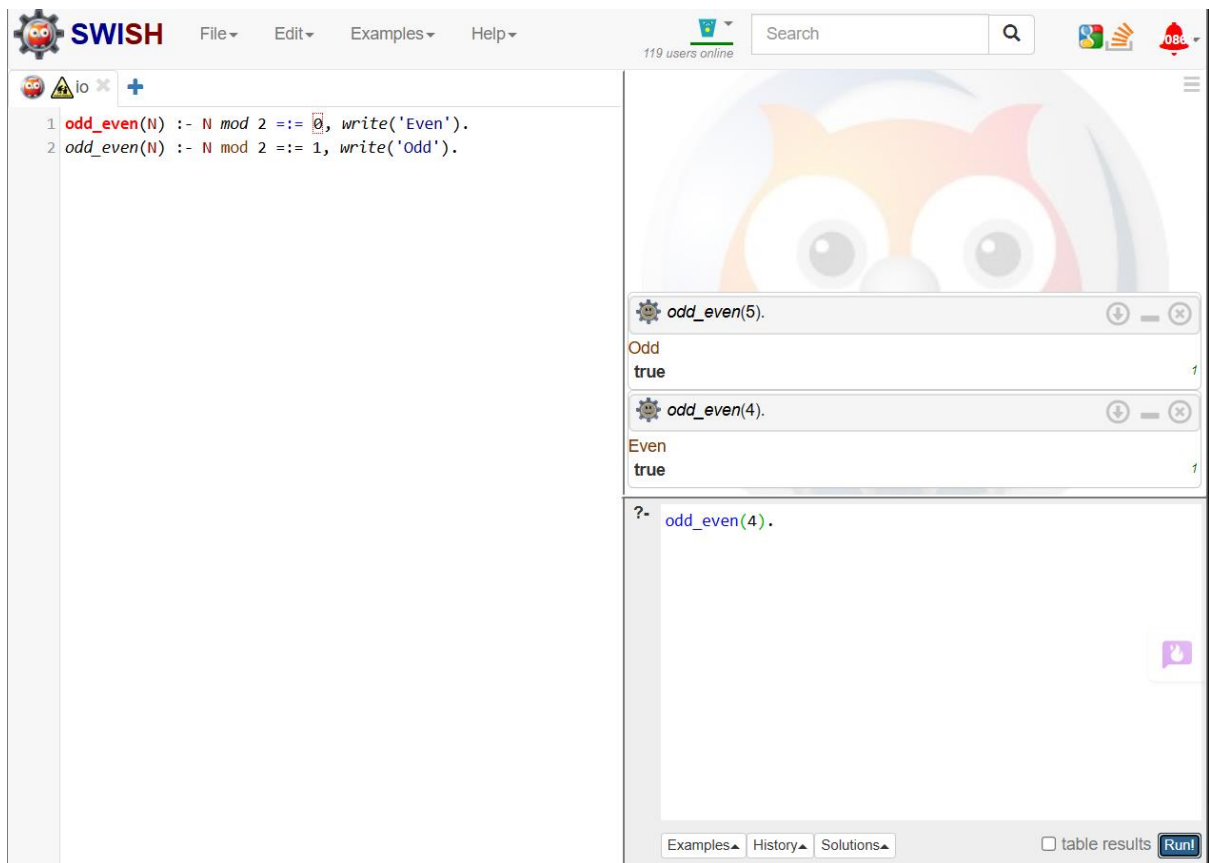
```
1 fact(0,1).
2 fact(N,F) :- N1 is N-1, fact(N1,F1), F is N*F1.
```

The console shows the results of queries:

- Query: `fact(0,1).` Result: `true`
- Query: `?- fact(0,1).` (Pending)

The interface includes a menu bar (File, Edit, Examples, Help), a search bar, and a status bar (116 users online, 086).

17.



The SWISH interface for problem 17 shows a code editor on the left with two lines of Prolog code:

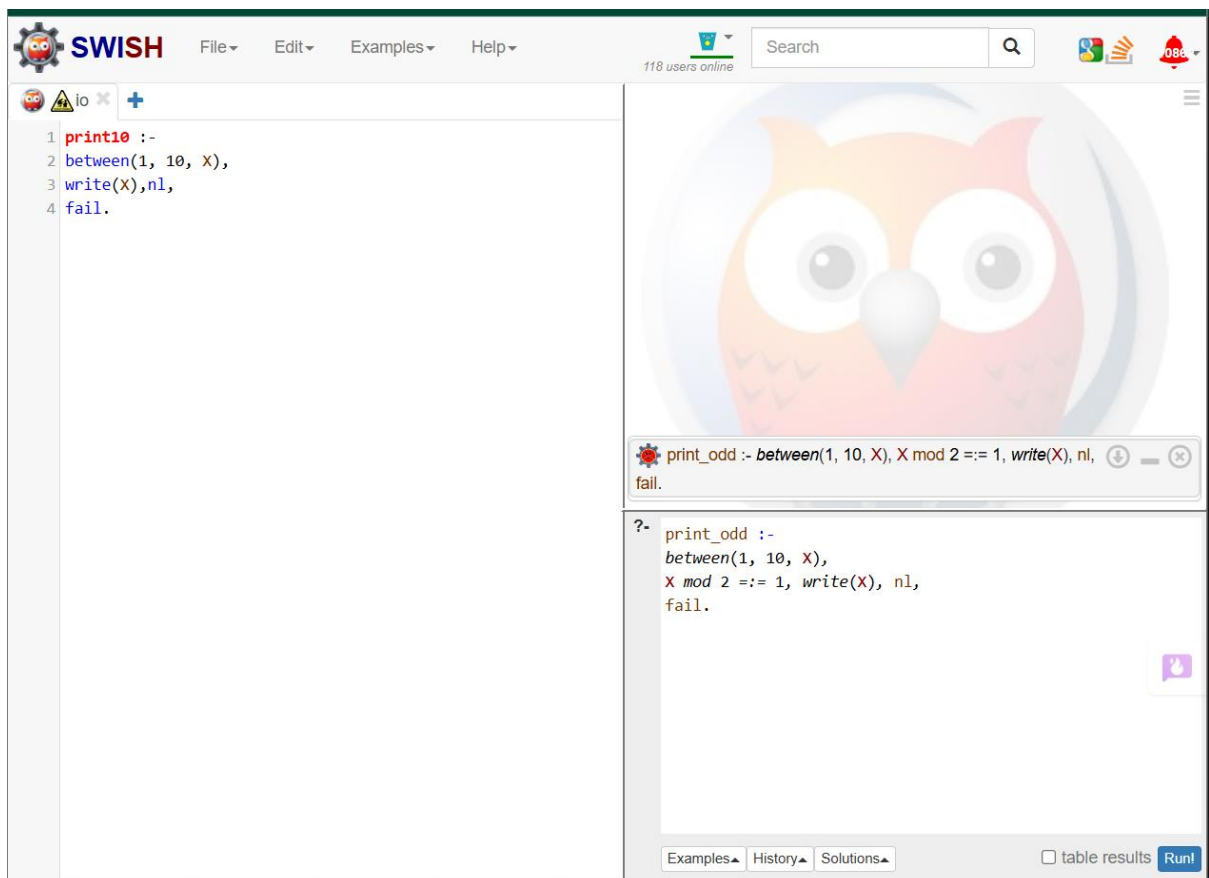
```
1 odd_even(N) :- N mod 2 =:= 0, write('Even').
2 odd_even(N) :- N mod 2 =:= 1, write('Odd').
```

The right panel features a large owl illustration. Below it, a console window displays the results of running the code:

```
odd_even(5).
Odd
true
odd_even(4).
Even
true
```

At the bottom of the right panel, there is a search bar with the text "odd_even(4).", a "table results" checkbox, and a "Run" button.

18.



The SWISH interface for problem 18 shows a code editor on the left with four lines of Prolog code:

```
1 print10 :-
2 between(1, 10, X),
3 write(X),nl,
4 fail.
```

The right panel features a large owl illustration. Below it, a console window displays the results of running the code:

```
print_odd :- between(1, 10, X), X mod 2 =:= 1, write(X), nl,
fail.
```

At the bottom of the right panel, there is a search bar with the text "print_odd :-", a "table results" checkbox, and a "Run" button.

19.

SWISH File Edit Examples Help 121 users online

```

1 male(john). male(peter). female(mary).
2 parent(john, peter). parent(mary, peter).
3 father(X, Y) :- male(X), parent(X, Y).
4 mother(X, Y) :- female(X), parent(X, Y).

```

parent(X,Y)

```

X = john,
Y = peter
X = mary,
Y = peter

```

male(X)

```

X = john
X = peter

```

mother(X,Y)

```

X = mary,
Y = peter

```

mother(mary, peter).

```

true

```

?- mother(mary, peter).

Examples History Solutions table results Run!

20.

SWISH File Edit Examples Help 123 users online

```

1 hanoi(1,A,_C) :- write(A), write(' -> '), write(C), nl.
2 hanoi(N,A,B,C) :-
3   N > 1, M is N-1,
4   hanoi(M,A,C,B), hanoi(1,A,B,C), hanoi(M,B,A,C).

```

hanoi(3,left,middle,right).

```

left -> right
left -> middle
right -> middle
left -> right
middle -> left
middle -> right
left -> right
true
false

```

?- hanoi(3,left,middle,right).

Examples History Solutions table results Run!